1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlights that immune-contexture readouts (Immunoscore®, immunotherapy biomarkers) do not uniformly predict outcomes in muscle-invasive bladder cancer (MIBC); why do some TIL-rich tumors fail to respond to immunotherapy?

Working Hypothesis

In muscle-invasive bladder cancer (MIBC) patients treated with immune checkpoint inhibitors (ICIs), high systemic HDL-C reconfigures the tumor lipid microenvironment such that abundant tumor-infiltrating lymphocytes (high Immunoscore) do not translate into effective anti-tumor immunity. This HDL-C–driven remodeling promotes immunosuppressive stromal and myeloid states (e.g., M2-like tumor-associated macrophages and PTGDS/prostaglandin signaling activity), leading to impaired CD8+ T cell effector function and reduced therapy_response. Consequently, the predictive value of Immunoscore alone is attenuated in the context of elevated HDL-C, revealing a HDL-C × Immunoscore interaction that stratifies responders from non-responders within high-TIL tumors. Secondary modifiers—SARIFA-driven tumor–adipocyte/stromal remodeling, PTGDS expression, and AI/radiomics features reflecting metabolic-immune crosstalk—will further refine prediction and provide noninvasive readouts of the underlying biology. Testable predictions include: (1) among Immunoscore-high tumors, higher HDL-C will associate with lower therapy_response rates to ICIs; (2) tumor tissues from HDL-C–high cases will show increased PTGDS expression, prostaglandin signaling activity, and M2-like macrophage markers with reduced CD8+ cytotoxic signatures; (3) in vitro and ex vivo models will demonstrate that HDL-containing media promote immunosuppressive phenotypes and blunt T-cell activation in the presence of bladder cancer–associated myeloid cells; (4) radiomic features tied to SARIFA/adipocyte remodeling will correlate with HDL-C levels and predict treatment outcome. This framework yields testable, clinically actionable insights into metabolic–immune crosstalk in MIBC.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	HDL-C, Immunoscore®
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

HDL-C: HDL-C path (introduces a plausible systemic/metabolic modifier).

Immunoscore®: Immunoscore®, immunotherapy biomarkers) do not uniformly predict outcomes in muscle-invasive bladder cancer (MIBC); why do some TIL-rich tumors fail to respond to immunotherapy? (2) Tier 1 paths used: I built on the Immunoscore®/immunotherapy-biomarker axis (shows that local T-cell contexture should predict immunotherapy benefit) and the HDL-C path (introduces a plausible systemic/metabolic modifier).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	SARIFA (tumor–adipocyte/stromal interaction)	Supporting modifier from knowledge graph
Secondary	PTGDS (prostaglandin D synthase / prostaglandin signaling)	Supporting modifier from knowledge graph
Secondary	AI-based biomarkers / radiomics-derived heterogeneity features	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- SARIFA (tumor–adipocyte/stromal interaction): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PTGDS (prostaglandin D synthase / prostaglandin signaling): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.